StoneNet

A Graph-Based Framework for Kidney Stone Drug Response Prediction

CSE 881 - Data Mining

Project Intermediate Report

Preeti Lakshmi Narayan

Hemprasanna Anbarasan

Caroline Newton

Bharath Mikkilineni

Abstract

This project develops **StoneNet** which is a graph neural network system to predict drug response for kidney stone (nephrolithiasis) patients. We collected clinical trial data from ClinicalTrials.gov by querying for kidney stone related studies. The raw dataset contained 324 trials across 11 attributes. After preprocessing, 323 trials had usable outcome data, with 58 of those having published results. The dataset covers 614 total intervention records across 11 categories, with 144 unique drug interventions identified. The cleaned data will be used to construct a heterogeneous knowledge graph connecting drugs, diseases, trial phases, and outcomes. A Relational Graph Convolutional Network (R-GCN) with a path-based attention mechanism will be trained on this graph to predict drug effectiveness. The model will be evaluated against two baseline methods and demonstrated through an interactive Streamlit prototype.

1 Difficulty Level

	Criteria	Description	Pts
(a)	Data Collection	Custom Python scripts to query the ClinicalTrials.gov REST API and parse the raw CSV. No pre-packaged dataset was used.	1.0
(b)	Data Preprocessing	Multi-step pipeline: text cleaning, intervention splitting, name normalization, outcome flag creation, and three output file exports, all written from scratch.	1.0
(c)	Algorithm Development	Implementing HPD-KGA, a novel R-GCN with a path-based attention mechanism. Not based on any existing published model.	2.0
(d)	Ground Truth Labels	Labels derived from trial outcome flags. Trials with posted results or outcome measures are treated as positive labels.	0.5
(e)	Evaluation	Model compared against Node2Vec and Matrix Factorization baselines using AUC-ROC, F1, and Precision-Recall.	0.5
(f)	Prototype	Interactive Streamlit interface with drug input and subgraph visualization using NetworkX.	1.0
		Total	**6.0**

Classification: **Hard — 6.0 points total.**

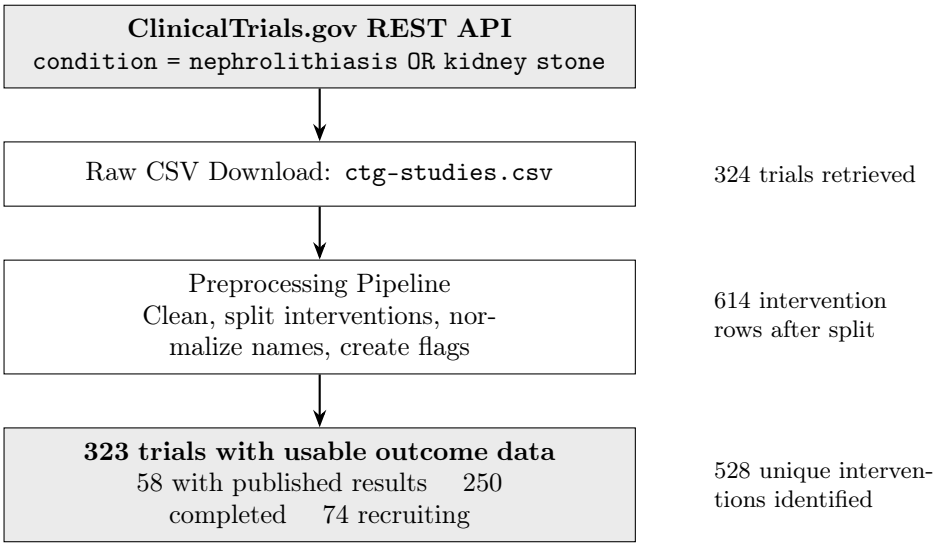
2 Data Collection

2.1 Source and Method

We used the ClinicalTrials.gov REST API to collect clinical trial records related to kidney stone treatment. The search used the condition terms **nephrolithiasis** and **kidney stone**. The data was downloaded as a single CSV file (`ctg-studies.csv`).

2.2 Data Collection Flow

The figure below shows how data moved from the API to the final filtered dataset.



2.3 Dataset Statistics

Metric	Value
Total trials retrieved	324
Trials with usable outcome data	323
Trials with published results	58
Completed trials	250
Recruiting trials	74
Total intervention rows	614
Unique interventions identified	528
Unique drug interventions	144
Average enrollment per trial	191 participants
Median enrollment per trial	86 participants

2.4 Attributes Collected

Each trial record includes the following attributes:

Attribute	Description
`nct_number`	Unique trial identifier
`study_title`	Full name of the trial
`study_status`	Completed / Recruiting / Terminated
`phases`	Trial phase (Phase 1 through Phase 4)
`enrollment`	Number of participants enrolled
`conditions`	Medical condition being studied
`interventions`	Drugs or procedures (pipe-separated in raw data)
`primary_outcome_measures`	Main endpoints measured
`secondary_outcome_measures`	Secondary endpoints measured
`study_results`	Whether results are posted (Yes / No)

2.5 Intervention Category Breakdown

After splitting the interventions column, the 614 rows fall into the following categories. The bar chart below shows the count per category.

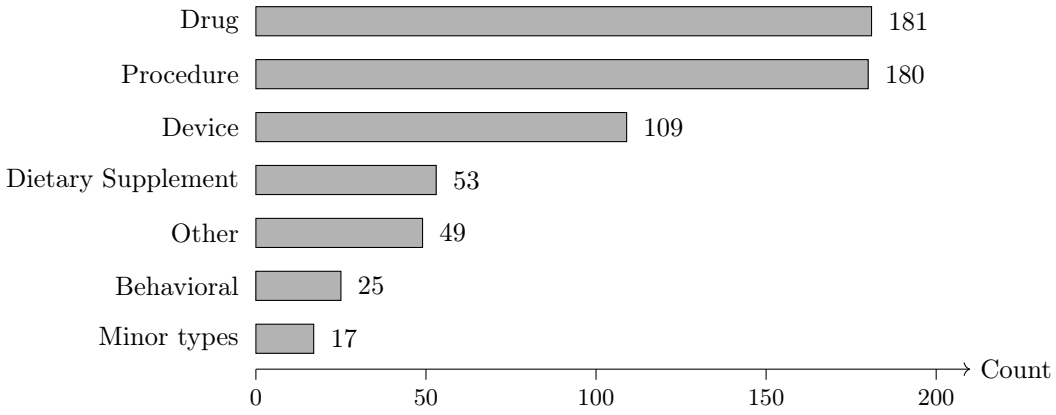


Figure 1: Intervention categories across 614 rows

2.6 Top Drug Interventions

The table below lists the most frequently appearing drug interventions:

Drug	Trials	Completed	With Results
Placebo	18	16	5
ALLN-177	4	4	1
Tamsulosin	4	4	1
Ketorolac	3	3	3
Morphine	3	3	0
Pregabalin	2	2	2
Tolvaptan	2	2	1
Ropivacaine	2	2	1

Note: Placebo appears most often because most trials are placebo-controlled.

2.7 Ground Truth Labels

A trial is labeled as usable (positive) if it meets at least one of the following:

- `study_results = Yes` — results have been officially posted
- `primary_outcome_measures` is not empty
- `secondary_outcome_measures` is not empty

This produced the `usable_outcome_data` binary flag. 323 out of 324 trials were labeled positive. Only 1 trial had no usable data at all.

3 Data Preprocessing

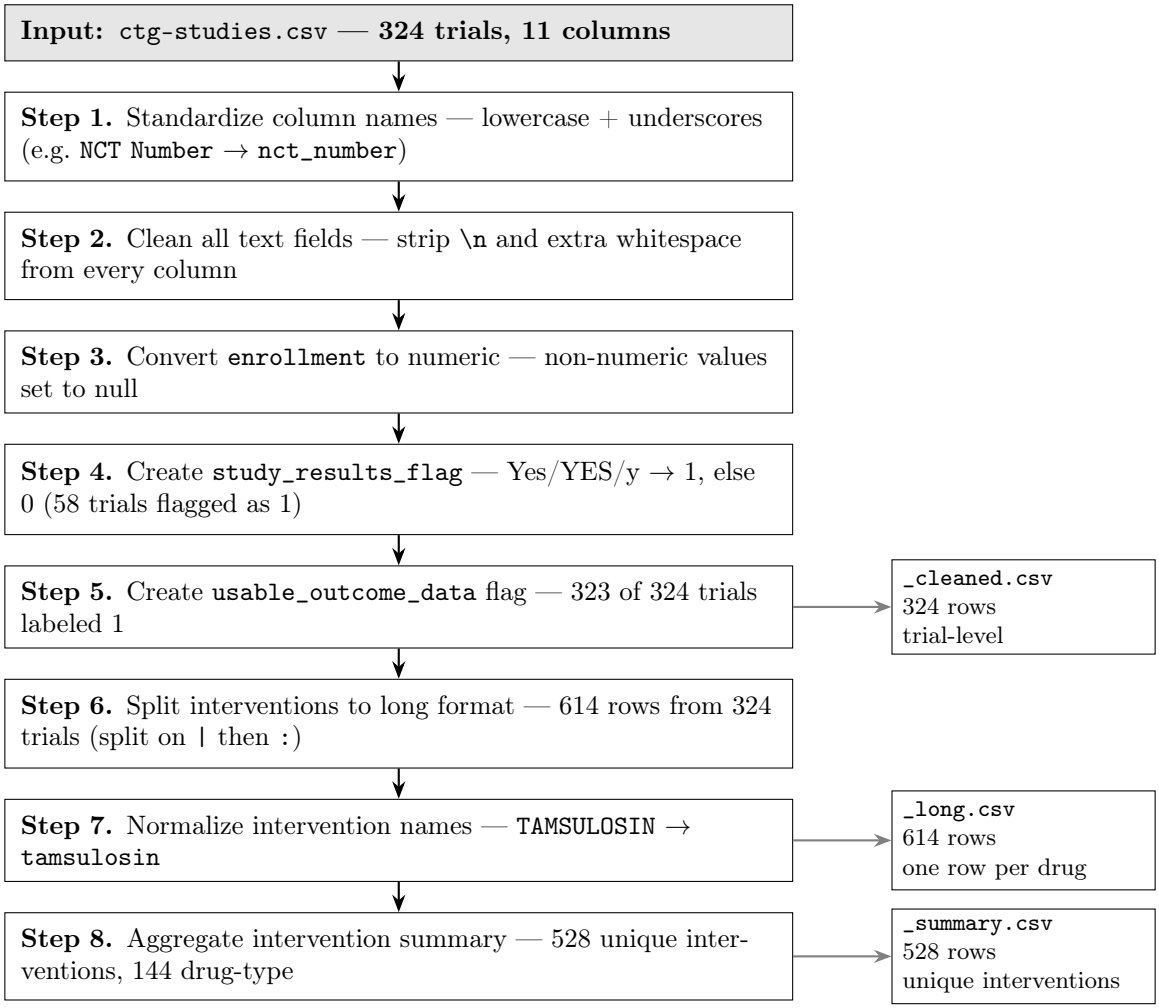
3.1 Issues Found in the Raw Data

When we first loaded `ctg-studies.csv`, we found several problems that needed to be fixed:

- Column names had mixed capitalization and spaces (e.g., `NCT Number`, `Study Status`)
- Text fields contained embedded `\n` characters and irregular whitespace throughout
- The `Interventions` column stored multiple items per trial as one pipe-separated string, for example: `DRUG: Tamsulosin | PROCEDURE: Ureteroscopy | DEVICE: Laser`
- The `Enrollment` column was stored as text with non-numeric values in some rows
- The `Study Results` column used inconsistent formats: `Yes`, `YES`, `y`, `NO`
- No label column existed for training — it had to be built from three separate columns
- The same drug appeared in different letter cases across rows (e.g., `Tamsulosin` vs. `TAMSULOSIN`)

3.2 Preprocessing Steps Completed

All preprocessing was done by Caroline using a custom Python script. The pipeline diagram below shows each step in order.



3.3 Output Files Produced

File	Rows	Contents
`clinicaltrials_kidney_stone_cleaned.csv`	324	Trial-level cleaned table with two new flag columns added
`clinicaltrials_interventions_long.csv`	614	One row per drug per trial; includes intervention type, name, and normalized name
`clinicaltrials_intervention_summary.csv`	528	One row per unique intervention with trial count, average enrollment, completion status, and result availability

3.4 Remaining Preprocessing Steps

The following steps must be completed before model training can begin:

- Build the knowledge graph using PyTorch Geometric `HeteroData` — nodes will be drugs, diseases, and trials; edges will represent intervention and outcome relationships
- Handle class imbalance for rare drug types using weighted sampling
- Create 70/15/15 train/validation/test splits ensuring no drug entity appears across more than one split
- Final check for duplicate edges and isolated nodes in the graph

3.5 Planned Knowledge Graph Structure

Once preprocessing is complete, the cleaned data will be assembled into a heterogeneous knowledge graph. The diagram below shows the node types and relationships we plan to build.

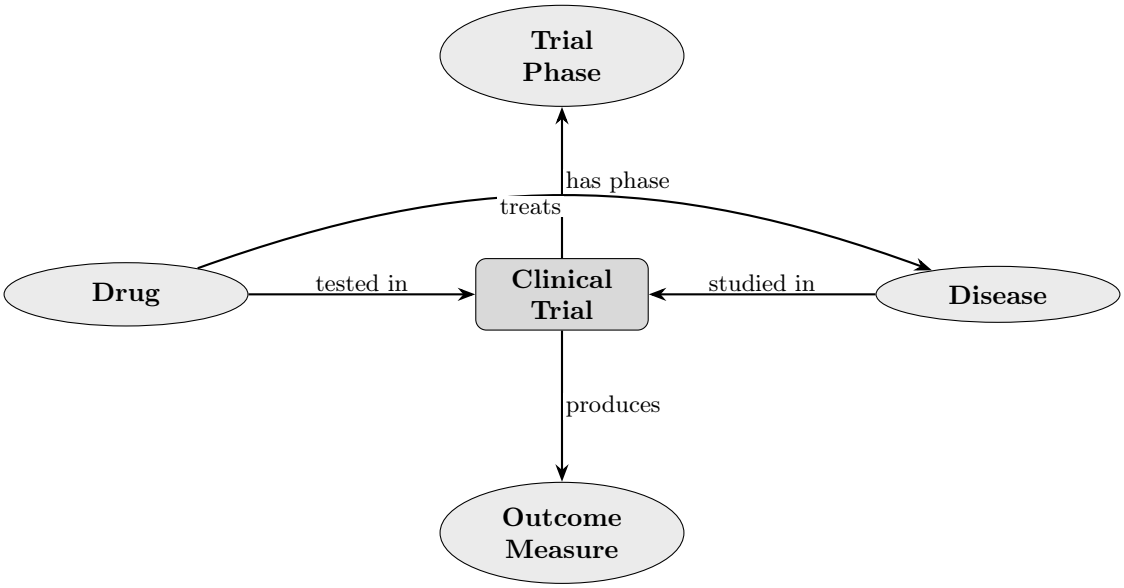


Figure 2: Planned knowledge graph structure

4 Timeline

Week 1: Knowledge Graph Construction

Team Members: Caroline, Preeti

In the first week, we will focus on building the knowledge graph from the cleaned dataset. Using PyTorch Geometric's **HeteroData** structure, we will organize different types of entities (such as drugs, diseases, etc.) and define how they are connected. The main goal here is to make sure the data is correctly represented as a graph, since this forms the foundation for everything that follows.

Week 1–2: R-GCN Encoder Implementation

Team Member: Preeti

Next, we will implement the R-GCN model, which is designed to work with graphs that have multiple types of relationships. Instead of directly applying it to the full dataset, we will first test it on a smaller portion of the graph. This helps us verify that the implementation is working correctly before scaling up.

Week 2: Path Extraction Module

Team Member: Caroline

At this stage, we will develop a module to extract paths between nodes in the graph. These paths represent indirect relationships (for example, connections between a drug and a disease through other entities). This step is important because these paths will later help the model capture deeper relationships beyond direct links.

Week 2–3: Path-Aware Attention and Integration

Team Members: Preeti, Caroline

We will then build a path-aware attention mechanism that allows the model to focus on the most important paths between entities. Not all connections are equally useful, so this component helps prioritize the meaningful ones. This attention mechanism will be combined with the R-GCN using a fusion layer, bringing together both structural and path-based information.

Week 3: Model Training and Tuning

Team Members: Preeti, Caroline

Once the model is ready, we will begin training it on the dataset. During this process, we will experiment with different settings such as learning rate, dropout, and the number of layers. The aim is to find a balance where the model performs well without overfitting.

Week 3: Repository Setup and Baseline Model

Team Member: Hemprasanna

Alongside model development, we will set up a GitHub repository to keep everything organized. A baseline model using Node2Vec will also be implemented. This gives us a simple reference point, so we can clearly compare how much improvement our main model provides using metrics like AUC and F1-score.

Week 3–4: Model Evaluation

Team Members: Preeti, Hemprasanna

After training, we will evaluate the model using standard metrics such as AUC-ROC, F1-score, and Precision-Recall. This step helps us understand how well the model is performing and whether it is reliable across different evaluation criteria.

Week 4: Streamlit Interface Development

Team Member: Bharath

We will also build a simple user interface using Streamlit. This will allow users to input a drug and see the model's predictions in a clear and interactive way. The focus here is usability, making the system easy to demonstrate and understand.

Week 4: Qualitative Path Analysis

Team Member: Caroline

In addition to quantitative evaluation, we will analyze the model's predictions qualitatively. This involves looking at the important paths the model used to make decisions. Understanding these paths helps us explain the model's behavior, which is especially important in a healthcare context.

Week 4–5: Final Report Writing

Team Members: All

All team members will contribute to writing the final report. The goal is to clearly document our approach, experiments, and findings in a structured and readable manner.

Week 5: Presentation and Demo Preparation

Team Members: Bharath, All

Finally, we will prepare presentation slides and practice the demo. This ensures that we can clearly communicate our work and confidently present both the technical details and the overall impact of the project.

5 Team Member Contributions

Team Member	Contributions
Preeti Lakshmi Narayan	**Past:** Designed the StoneNet graph schema. Defined node types (drugs, diseases, trials, outcomes) and edge types based on the ClinicalTrials.gov data structure. **Future:** Lead graph construction in PyTorch Geometric. Implement R-GCN encoder. Build path-aware attention mechanism. Lead model training and tuning. Write algorithm and methodology sections of the final report.
Caroline Newton	**Past:** Queried the ClinicalTrials.gov API and downloaded the raw CSV (324 trials, 11 attributes). Wrote the complete 8-step preprocessing pipeline: standardized columns, cleaned text, fixed enrollment, created outcome flags, split 614 intervention rows into long format, normalized drug names, and produced the three output files. **Future:** Assemble the PyG HeteroData graph. Implement the path extraction module. Run qualitative analysis on predictions. Write the data and preprocessing sections of the final report.
Hemprasanna Anbarasan	**Future:** Set up the shared GitHub repository. Run the Node2Vec baseline using PyTorch Geometric and record AUC/F1 scores. Support model training by monitoring loss curves. Compile evaluation results into a summary table. Write the evaluation section of the final report.
Bharath Mikkilineni	**Future:** Build a Streamlit UI where users enter a drug name and see the predicted response score with a basic subgraph view using NetworkX. Prepare the class presentation slides. Write the prototype section of the final report.

AI Usage Citation: ChatGPT-5 (02/06/2026) was used for grammar refinement, LaTeX structuring and report drafting. All analysis and implementaion is our own idea, work and understanding.